

PilotFish Gaps, Bottlenecks & Custom Module Plan

Scoped to the Brian Wolfe workflow: **SFTP poll EDI 835** → **split each ST** → **JSON** → **Oracle OCI Object Storage**. This document is intentionally critical of product gaps the demo hits, and recommends **custom Java eiPlatform modules**.

Companion runnable demo: `Clients/Demos/edi-835-oci-bucket/`. See also *DESIGN.md*.

1. Target architecture (boss conversation)

SFTP Polling (EDI 835 files) → Split out each ST segment and create a JSON file → Send JSON to Oracle OCI bucket (using HTTP POST right now, since we do not have an OCI Transport).

2. Issue register

Sev	Gap / flaw / bottleneck	Why it matters	Recommendation
Critical	No native OCI Object Storage Transport (or Listener)	AWS S3 / GCS modules exist on 23R1; OCI does not. Forces HTTP workarounds and weak ops (no native opc-* headers, multipart, PAR).	Build <code>OciObjectStorageTransport</code> (Java) using OCI Java SDK or signed REST.
High	<code>HttpPostTransport</code> is POST-only	OCI <code>PutObject</code> is HTTP PUT to <code>/n/{ns}/b/{bucket}/o/{object}</code> . Method mismatch and wrong signing model.	Custom transport with configurable PUT/POST + upload API; or pre-authenticated request (PAR) URL mode.
High	No OCI request signing helper	Oracle requires tenancy/user/fingerprint + private key (API key) or instance principals / resource principals. Naked HTTP will 401 against real OCI.	Module config: region, namespace, bucket, auth mode (API key / instance / resource / PAR).
Med	ST split is multi-module choreography	Demo uses <code>EDI→XML + XPath //Transaction</code> . Fragile if Friendly Names / legacy XML / version tables change. Alternative <code>FileRecordForkingListener</code> / <code>EdiForkingModule</code> still not "835-aware JSON".	Optional <code>EdiStSplitToJsonProcessor</code> : ST/SE aware splitter emitting one JSON TX per ST with ISA/GS wrappers.
Med	Partial failure on multi-ST files	N HTTP calls after fork; mid-file OCI failure leaves half uploaded without a compensation store.	Transport batch + transaction store; or join + zip multipart upload.
Med	No first-class OCI observability	Missing <code>opc-request-id</code> correlation attributes, bucket/object attrs, eTag persistence.	Populate TX attrs from response headers in custom transport.
Low	Demo OCI mock ≠ IAM / CMEK / lifecycle	Accepted for lab; real customers need compartment policies, encryption keys, retention.	Document SDK module features; do not pretend mock == OCI.

3. Recommended custom Java modules

3.1 OciObjectStorageTransport (priority #1)

FQCN sketch: `com.pilotfish.eip.modules.oci.OciObjectStorageTransport` extends `AbstractTransport`. Categories: WEB, CLOUD, SPECIAL_PROTOCOL.

- Config: Region, Namespace, Bucket, ObjectName (OGNL), ContentType, AuthMode, ConfigFile/Profile or API key fields, Timeout, Multipart threshold.
- Behavior: PutObject (default), optional Get/Head for response listeners; set attributes `com.pilotfish.oci.requestId`, `eTag`, `opc-content-md5`.
- Deps: `oci-java-sdk-objectstorage` (+ common + bom). Ship as `modules-oci.jar` into WEB-INF/lib.
- Why not HttpPost: wrong verb, no signing, no multipart, no native PAR/SSE-C/KMS knobs.

Skeleton in this demo

`custom-modules/oci-object-storage-transport/` — compile against `eiPlatform` extend APIs + OCI SDK. Not loaded into the demo image until built/signed per customer license policy.

```
Key methods: getType()/getDescription(), getConfigurationDescriptor(), processTransaction(...) →
ObjectStorageClient.putObject(...).
```

3.2 EdiStSplitToJsonProcessor (priority #2)

- Input: raw X12; detect element / segment / repetition delimiters from ISA.
- Split on ST-SE boundaries; wrap each with reconstructed ISA/GS/GE/IEA (control # rewrite).
- Emit JSON ({documentType, controlNumber, segments[], monetarySummary}) or fork child transactions.
- Reduces XPath fragility vs EDITransformationModule + Friendly Names variance.

3.3 Optional OciObjectStorageListener

Poll bucket prefixes / events for downstream 835/999/277. Useful later; not required for this upload-only boss flow.

4. What the demo deliberately fakes

- Local Flask mock implements OCI-shaped paths and accepts PUT **and** POST.
- No OCI tenancy, compartments, IAM policies, or customer-managed keys.
- HttpPostTransport stands in for the missing OCI Transport (same workaround Brian described).
- Synthetic 2×ST 835 — not a certified SNIP clinic.

5. Suggested delivery plan for PF engineering

- Sprint A: `OciObjectStorageTransport` PutObject + API-key auth + attribute echo (parity with `AwsS3Transport`).
- Sprint B: PAR URL mode + multipart + retries with `opc-client-request-id`.
- Sprint C: Instance/resource principal auth for OKE/Compute.
- Sprint D (optional): `EdiStSplitToJsonProcessor` productized from this demo's XSLT logic.

6. Bottom line for stakeholders

PilotFish can already own the **SFTP** → **EDI** → **fork** → **JSON** half cleanly. The structural product hole is **Oracle Cloud Object Storage as a first-class Transport**. Continuing to paper over it with `HttpPostTransport` creates signing, method, multipart, and ops blind spots. Invest in a custom (then productized) Java **`OciObjectStorageTransport`**.